

La Nébuleuse

Sustainable Logistics Optimization

Vehicle Routing Problem & Facility Location Problem

Sustainable Logistics

Group 1

Hadrien Bischof
Valentine Krafft
Audrey Mathieu
Liam Trillo
Tifani Yazbeck

Executive Summary

This report provides a quantitative analysis aimed at optimizing the logistics network of the craft brewery La Nébuleuse to support its ongoing growth. By leveraging operations research methods, the study addresses three strategic pillars: vehicle routing (VRP), facility configuration (FLP), and fleet electrification.

Key Findings and Recommendations:

- **Routing Optimization (VRP):** The recommended Scenario establishes a standardized weekly delivery plan that balances legal compliance with operational efficiency at a cost of 269.20 CHF/week. This model optimally utilizes the heterogeneous fleet, comprising a VW Caddy, a Ford Transit, and a heavy truck, while maintaining a cost premium of less than 3 CHF/week compared to a theoretical, unconstrained optimum.
- **Infrastructure Strategy (FLP):** The analysis identifies a significant structural inefficiency caused by the 3.5 km separation between the Renens and Crissier sites, costing approximately 202 CHF/week in "first-mile" shuttling. The priority recommendation is to consolidate operations at the Renens site as soon as leasing opportunities arise, which would generate annual savings exceeding 10,500 CHF and simplify charging for future electric vehicles.
- **Energy Transition (EV):** While immediate electrification of urban vans is not profitable based solely on current VRP mileage, it becomes financially viable if total annual usage exceeds 15,000 km for the Caddy. The report advises adopting electric vehicles during the next replacement cycle if total mileage audits confirm these thresholds. Conversely, the electrification of the heavy truck should be deferred until 2028–2030 due to a prohibitive 265,000 CHF CAPEX premium.
- **Resilience and Growth:** The current urban fleet possesses sufficient capacity to absorb up to 40% growth in HORECA demand before requiring additional investment or a second daily tour.

In summary, the immediate deployment of the recommended routing plan, combined with a strategic consolidation at Renens, will allow La Nébuleuse to reduce its environmental footprint and operational costs while securing its logistics for future expansion.

Table of Contents

INTRODUCTION

1. Introduction & Context	3
1.1 Modelling Assumptions	3

COMPANY & DATA

2. Network Description & Demand Data	4
2.1 Network Infrastructure (Facilities)	4
2.2 Fleet Characteristics	4
2.3 Client Demand Profile	4
2.4 Data Cleaning & Weekly Demand Calculations	5
2.5 Separation of Logistics Flows.....	5

VRP — VEHICLE ROUTING PROBLEM

3. Mathematical Formulation (HFVRP)	7
3.1-3.4 Problem Class	7
3.5-3.7 Sets, Variables, Constraints, OR-Tools.....	8
4. VRP Routing Scenarios	9
4.1 Scenario 1 - Unrestricted Baseline.....	9
4.2 Scenario 2 - Strict Urban Exclusion (Infeasible).....	9
4.3 Scenario 3 - Flexible Allocation	10
4.4 Scenario 4 - Forced Lakeside Assignment.....	10
4.5 Scenario 5 - Final Operational Model (Recommended)	11
4.6 Scenario Comparison Summary	11

SENSITIVITY ANALYSIS

5. Sensitivity Analysis	12
5.1 Demand Growth & Fleet Resilience	12
5.2 Vehicle Operating Cost Sensitivity	12
5.3 Environmental Impact & CO ₂ Emissions.....	13

FACILITY LOCATION PROBLEM

6. Facility Location Problem (FLP)	14
6.1 Problem Definition & Infrastructure Bottlenecks.....	14
6.2 Objective of the Analysis	14
6.3 Hub Candidate Data	14
6.4 Mathematical Formulation (Multi-Echelon MILP).....	15
6.5 Results.....	15
6.6 Managerial Interpretation & Final Recommendation.....	16

ELECTRIC VEHICLE ANALYSIS

7. Electric Vehicle Fleet Analysis.....	17
7.1 Parameters & Assumptions	17
7.2 10-Year Total Cost of Ownership.....	17
7.3 Lifecycle CO ₂ Analysis	18
7.4 Sensitivity Analysis — Annual Mileage	18
7.5 Energy Price Sensitivity.....	19
7.6 Side Analysis — Heavy Truck	19
7.7 Final EV Recommendation	19

CONCLUSIONS

8. Recommendations	20
9. Conclusion & Future Extensions	21

1. Introduction & Context

La Nébuleuse is an independent craft brewery based in the Lausanne metropolitan area (Switzerland), producing and distributing a diverse range of beers, including kegs, bottled beer, and canned beer, to a network of bars, restaurants, and distributors across the region. As the company grows, the complexity of its logistics operations grows alongside it: more clients, heavier weekly volumes and a fleet of three very different vehicles that each serve distinct roles.

This report applies quantitative optimization methods to two interconnected logistics challenges facing La Nébuleuse:

- **Vehicle Routing Problem (VRP):** Optimizing current weekly delivery routes to minimize transportation costs, considering a heterogeneous fleet composed of a VW Caddy, a Ford Transit, and a heavy truck.
- **Facility Location Problem (FLP):** Evaluating three distinct strategic infrastructure scenarios: Status Quo, Production Consolidation in Renens, and a secondary hub in Givisiez, using Mixed Integer Programming (MIP) to provide a data-driven recommendation for long-term network design.

Both analyses are grounded in real operational data provided by La Nébuleuse management. The results are intended to serve as a practical decision-aid tool, quantitatively demonstrating that the choices made in vehicle assignment, route structure, and facility location have a measurable impact on both the financial efficiency and the environmental footprint of the company.

1.1 Modelling Assumptions

Before presenting the models, it is important to state the key simplifications and assumptions made to keep the analysis tractable while remaining operationally realistic:

- **No Time Windows:** Deliveries are scheduled without strict time-of-day constraints. While urgent (next-day) orders represent approximately 5% of La Nébuleuse's weekly operations, the vast majority of clients (95%) are flexible within the week.
- **Reverse Logistics Absorbed:** The collection of empty kegs and pallets during delivery stops is treated as a concurrent activity, effectively absorbed into the primary routing without generating additional trips or demand weight in the model.
- **Weekly Demand Snapshot:** Annual or multi-month sales volumes were divided by the number of active operational weeks per client to extract a representative average weekly demand. Two newer clients were adjusted: Simplon (12 weeks active) and La Latina (26 weeks active).
- **Separation of Logistics Flows:** Bulk tank ("Cuv") deliveries are handled separately by the Ford Transit operating from Brasserie Renens, as they use an incompatible vehicle format and originate from a different facility. These are excluded from the main VRP model.
- **Real Road Distance Matrix:** Distances between all nodes were compiled into a 14×14 distance matrix using actual road routing data, ensuring the model reflects real driving conditions rather than straight-line approximations.

2. Network Description & Demand Data

Before feeding data into the optimization models, it is critical to define the infrastructure, fleet capabilities, and demand profiles shaping La Nébuleuse's logistics. This section describes each element of the network and explains how raw sales data (extracted from the company's ERP system and provided directly by management) was processed into the standardized weekly demand figures used by the VRP and FLP models.

2.1 Network Infrastructure (Facilities)

The company operates across two main logistical nodes:

- **Brasserie Renens (Production):** The primary brewing and packaging site, located at Ch. du Closel 1, Renens. Covering approximately 1,200 m² of production and packaging space, it has limited storage capacity (~150 pallet spaces). This site also serves as the origin for bulk tank ("Cuv") deliveries made directly by the Ford Transit.
- **Entrepôt Crissier (Distribution Depot):** The dedicated logistics warehouse at Ch. des Lentillières 2, Crissier. Covering approximately 1,000 m² with ~500 pallet spaces, it serves as Node 0 (the Depot) in the VRP: the point where all delivery vehicles start and end their routes each week.

The 3.5 km separation between these two sites is a structural inefficiency: all finished goods must be shuttled from Renens to Crissier before being dispatched, generating unnecessary internal transport costs and handling. This is addressed in the FLP analysis (Section 6).

2.2 Fleet Characteristics (Heterogeneous Fleet)

Distribution is handled by a mixed fleet of three vehicles, each with distinct capacities and operational profiles, requiring a Heterogeneous Fleet VRP (HFVRP) approach where the objective minimizes total weekly CHF cost rather than distance alone:

Vehicles	Max Payload	Cost / km	Primary Use	Urban Access
VW Caddy	700 kg	0.60 CHF/km	Urban bar deliveries	✓ Excellent
Ford Transit	1,200 kg	0.80 CHF/km	Urban multi-stop / Cuv tanker	✓ Good
Heavy Truck	8,300 kg / 15 pal.	1.50 CHF/km	Bulk long-haul delivery	✗ Unsuitable

2.3 Client Demand Profile

La Nébuleuse serves 12 delivery clients split into four categories:

A. Urban HORECA Clients (City Centre)

Great Escape, Pointu, Simplon, Loxton, La Latina, Doki Doki, located in dense Lausanne streets, ordering 49–280 kg/week of packaged beer. The heavy truck cannot safely navigate this environment.

B. Lakeside Clients (Peripheral Zone)

The Lacustre, Migros Ouchy, Jetée de la compagnie and La Rincette along the lake waterfront. The truck passes near these clients on its way to Alloboissons, making them natural candidates for truck servicing. Jetée de la compagnie also receives Cuv bulk deliveries.

C. Dual-Flow Client

Docks (Av. de Sévelin 34) the largest urban packaged client (550 kg/week) plus regular Cuv bulk deliveries (259 L/week). The VW Caddy handles the packaged goods; the Ford Transit handles the bulk flow from Renens.

D. Bulk Distributor

Alloboissons (Givisiez, ~80 km from Crissier) 5,385 kg/week (72% of total VRP demand), fully palletized, requires the truck, and demands a dedicated direct highway run.

2.4 Data Cleaning & Weekly Demand Calculations

All client orders were converted from mixed ERP units (kegs, box of bottles, box of cans, bulk litres) to kilograms using the following unit weights and palletisation rules provided by La Nébuleuse:

Product Unit	Weight	Palletisation Rule	Pallet Tare
1 Keg (fût 20L)	26 kg	1 pallet = 30 kegs + 2 wooden pallets	40 kg
1 Box of Bottles (24×)	14 kg	1 pallet = 54 cartons + 1 wooden pallet	20 kg
1 Box of Cans (24×)	12 kg	1 pallet = 63 cartons + 1 wooden pallet	20 kg
Bulk / Cuv (tanker)	1 kg/L	No pallets, direct tanker flow	N/A

Processing steps: **(1) Filter out deposits** ("Caution") and negative returns. **(2) Categorize by ERP unit code**: Kegs (F), Bottles (6X4B, 24B...), Cans (6X4C...), Bulk (L). **(3) Divide annual volumes by active operational weeks**: 52 weeks standard, 12 weeks for Simplon, 26 weeks for La Latina (partially from manual field data). Values are integer-truncated (.astype(int)) to match OR-Tools integer inputs.

Verified weekly demand (from données2025.xlsx, replicating the notebook pipeline):

#	Client	Op. Weeks	VRP Demand (kg/week)	Bulk Flow (L/week)	Scenario 5 Vehicle
0	Brasserie Renens (Production)	-	0	-	(production site)
1	Entrepôt Crissier (VRP Depot)	-	0	-	(depot)
2	Great Escape	52	215	-	Ford Transit
3	The Lacustre	52	169	-	Truck (lakeside)
4	Simplon	12	280	-	Ford Transit
5	Pointu	52	80	-	Ford Transit
6	Loxton	52	49	-	Ford Transit
7	La Latina	26	80	-	Ford Transit
8	Docks	52	550	259	VW Caddy
9	Doki Doki	52	128	-	Ford Transit
10	Jetée de la compagnie	52	426	417	Truck (lakeside)
11	La Rincette	52	42	-	Truck (lakeside)
12	Migros Ouchy	52	78	-	Truck (lakeside)
13	Alloboissons (Givisiez)	52	5385	-	Truck (bulk)

Colour	Meaning
White / light grey	Standard urban HORECA clients
Light green	Lakeside clients: truck picks up on return from Alloboissons
Light yellow	Dual-flow clients: packaged goods (VRP) + Cuv bulk tanker (Ford Transit from Renens)
Light red	Alloboissons: bulk distributor, dedicated truck highway run

Orange numbers in the Bulk Flow column indicate a separate Cuv tanker delivery in addition to the standard packaged goods route.

Total VRP demand: 7,482 kg/week across all 12 clients

- Urban HORECA + Lakeside: 2,097 kg/week
- Alloboissons: 5,385 kg/week (72% of total).

2.5 Separation of Logistics Flows (Packaged vs. Bulk)

Bulk Cuv tanker deliveries and standard packaged goods originate from different facilities and require incompatible vehicles. They are modelled as two independent parallel flows:

Flow	Origin	Vehicle(s)	Metric	Routing Type
Standard VRP (Packaged goods)	Entrepôt Crissier	VW Caddy + Truck	kg / week	Multi-stop optimisation
Bulk Tank flow (Cuv)	Brasserie Renens	Ford Transit	Litres / week	Direct round trips

The main VRP model (Sections 3-5) covers exclusively the Standard Flow from Crissier. Clients with both flows (Docks and Jetée de la compagnie) appear in both streams simultaneously.

3. Mathematical Formulation (HFVRP)

The distribution problem is formally modelled as a Heterogeneous Fleet Vehicle Routing Problem (HFVRP), extending the classical Capacitated VRP in two ways: a heterogeneous fleet (different capacity and cost per km per vehicle) and a cost minimization objective (not distance). The model is solved using Google OR-Tools.

3.1 Problem Class

- **Heterogeneous fleet:** Each vehicle has a different payload capacity and cost per kilometre, so the solver must decide not only which clients each vehicle visits, but also which vehicle type is most economical for each route.
- **Cost minimization (not distance):** The objective function minimizes total weekly CHF cost. The truck costs 1.50 CHF/km (2.5× more than the Caddy) so sending it into the city for small drops is financially penalized even if the distance would be short.

3.2 Sets and Indices

$N = \{0, 1, \dots, n\}$: All nodes, where 0 is the depot (Entrepôt Crissier).

$C = \{1, \dots, n\}$: All client nodes to be visited.

$V = \{1, \dots, K\}$: Available vehicles ($K = 3$: VW Caddy, Ford Transit, Truck). Note: the notebook's written formulation refers to 'Caddy, Truck' but the code implements all three.

$i, j \in N$: Indices for nodes (origin and destination).

$k \in V$: Index for vehicles.

3.3 Parameters

c_{ij} : Road distance in metres between node i and node j (14×14 distance matrix).

α_k : Variable cost per kilometre for vehicle k (0.60 / 0.80 / 1.50 CHF/km).

d_i : Weekly demand of client i in kg (integer-truncated from processed Excel data).

Q_k : Maximum weight capacity of vehicle k (700 / 1,200 / 8,300 kg).

P_k : Maximum pallet capacity of vehicle k (∞ for vans | 15 pallets for Truck).

3.4 Decision Variables

$x_{ijk} \in \{0,1\}$: Binary: 1 if vehicle k travels directly from node i to node j .

$u_{ik} \geq 0$: Continuous auxiliary: cumulative load of vehicle k after visiting node i (MTZ subtour elimination).

3.5 Objective Function

Minimize the total weekly transportation cost across all vehicles and arcs:

$$\min Z = \sum_{(k \in V)} \sum_{(i \in N)} \sum_{(j \in N)} \alpha_k \cdot c_{ij} \cdot x_{ijk}$$

Minimize total cost (CHF/week) = vehicle cost rate × distance × whether arc is used

3.6 Constraints

- **A: Customer Coverage:** Every client visited exactly once by exactly one vehicle: $\sum_{k \in V} \sum_{j \in N, j \neq i} x_{ijk} = 1 \quad \forall i \in C$
- **B: Flow Conservation:** If vehicle k arrives at node j , it must also depart: $\sum_i x_{ijk} - \sum_l x_{jlk} = 0 \quad \forall j \in C, \forall k \in V$
- **C: Depot Continuity:** Each vehicle leaves the depot at most once: $\sum_{j \in C} x_{0jk} \leq 1 \quad \forall k \in V$
- **D: Vehicle Capacity:** Total weight per vehicle cannot exceed payload: $\sum_{i \in C} d_i \cdot \sum_{j \in C} x_{ijk} \leq Q_k \quad \forall k \in V$
- **E: Subtour Elimination (MTZ):** $u_{ik} - u_{jk} + Q_k \cdot x_{ijk} \leq Q_k - d_j \quad \forall i, j \in C, i \neq j, \forall k \in V$, prevents isolated sub-loops not including the depot.

3.7 OR-Tools Implementation

The model is implemented using Google OR-Tools' pywrapcp routing engine with these key settings:

OR-Tools Parameter	Value / Setting
First solution strategy	PATH_CHEAPEST_ARC: builds initial route greedily by cheapest next arc
Local search metaheuristic	GUIDED_LOCAL_SEARCH: escapes local optima by penalising frequent arcs
Time limit	10 seconds per scenario
Arc cost function	Distance (m) $\times$ cost per metre for that vehicle: minimises total CHF
Capacity dimension	Weight (kg) per vehicle, hard upper bound, cumul starts at 0

Implementation detail: distances are converted to integer metres ($\times 1,000$). Arc cost = (distance_metres $\times$ cost_centimes_per_km) / 1,000, keeping all values as integers while preserving precision.

4. VRP Routing Scenarios

This section presents the five progressive scenarios developed to find the optimal, operationally realistic weekly routing plan. Each scenario builds on the lessons of the previous one, progressively introducing real-world constraints until the final recommended solution is reached.

Scenario 1 : Unrestricted Baseline

INFEASIBLE IN PRACTICE

Setup & Result

No geographic restrictions. The solver assigns all deliveries to the truck (Sweep Effect): using one vehicle avoids the fixed distance cost of dispatching a second vehicle from the depot. Result: all 12 clients served by the truck in one loop.

Vehicle	Route (stop order)	Load	Distance	Cost
VW Caddy	Crissier → Crissier (idle)	0 kg	0 km	0.00 CHF
Ford Transit	Crissier → Crissier (idle)	0 kg	0 km	0.00 CHF
Truck	Crissier → [all 12 clients incl. Alloboissons] → Crissier	~7,482 kg	~160 km*	~241 CHF*
TOTAL		~7,482 kg	~160 km*	~241 CHF*

* Approximate.

Key finding is structural: all load on truck, vans idle. An 8,300 kg truck navigating narrow Lausanne streets for 80 kg bar deliveries is operationally and legally impossible.

Scenario 2 : Strict Urban Exclusion

NO SOLUTION FOUND

Setup & Result

Truck forbidden from all urban nodes (nodes 2-12). The solver returns: "No solution found!" This is a critical operational insight, not a bug:

- **Total urban + lakeside demand:** ~2,097 kg
- **Total van capacity:** 700 + 1,200 = 1,900 kg (197 kg short of total demand)

The combined van capacity is physically insufficient. Management must either allow the truck at peripheral (lakeside) nodes or schedule afternoon reload tours.

Scenario 3 : Flexible Allocation**OPERATIONAL ✓****Setup & Result**

Fleet expanded to 5 virtual vehicles (Tour 1 & 2 for vans). Truck allowed at lakeside nodes but not dense city centre. Weekly cost: 266.39 CHF | 196.43 km.

Vehicle	Route (stop order)	Load	Distance	Cost
VW Caddy	Crissier → Loxton → La Latina → Docks → Crissier	679 kg	~15 km	~9 CHF
Ford Transit	Crissier → Pointu → Great Escape → Doki Doki → Simplon → Jetée → La Rincette → Crissier	1,171 kg	~18 km	~15 CHF
Truck	Crissier → Alloboissons → The Lacustre → Migros Ouchy → Crissier	5,632 kg	~163 km	~242 CHF
TOTAL		7,482 kg	196.43 km	266.39 CHF

Near-perfect utilization: Caddy 679/700 kg (97%), Transit 1,171/1,200 kg (97.5%). The truck naturally sweeps lakeside clients on its return from Alloboissons.

Scenario 4 : Forced Lakeside Assignment**OPERATIONAL ✓****Setup & Result**

As the VRP does not account for the actual road network, it cannot recognize that when the truck drives to Migros Ouchy and The Lacustre, it passes near La Rincette and the Jetée de la Compagnie. To improve the realism of Scenario 3, which relies on Euclidean “as-the-crow-flies” lines to connect nodes, we forced the truck to serve these two locations as well in the final model. The lakeside nodes (Migros Ouchy, The Lacustre, Jetée, La Rincette) were pinned to the truck using SetValue.

Cost difference vs Scenario 3: just 0.25 CHF/week, validating the solver's original decision.

Vehicle	Route (stop order)	Load	Distance	Cost
VW Caddy	Crissier → Loxton → La Latina → Docks → Crissier	679 kg	~15 km	~9 CHF
Ford Transit	Crissier → Pointu → Great Escape → Doki Doki → Simplon → Jetée → La Rincette → Crissier	1,171 kg	~18 km	~15 CHF
Truck	Crissier → Alloboissons → The Lacustre → Migros Ouchy → Jetée → La Rincette → Crissier	6,054 kg	~163 km	~242 CHF
TOTAL		7,482 kg	196.43 km	266.64 CHF

Scenario 5: Final Operational Model, Forced Sequence

RECOMMENDED

Setup & Result

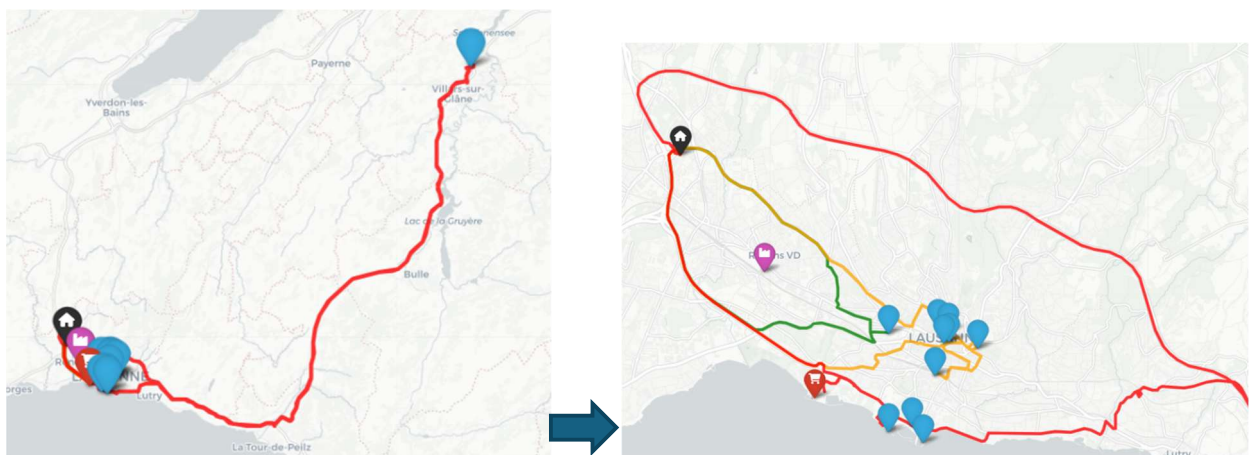
The tour created by the algorithm makes the truck first visit the nearest node (La Rincette). However, it could be more optimal to modify the sequence by visiting Alloboisson first. This would make the truck lighter for the rest of the tour, potentially reducing CO₂ emissions and fuel consumption. To better align the mathematical optimization with human operational logic, we forced the algorithm to choose Alloboisson as the first node to visit. This was implemented by forcing `NextVar(truck_start) = alloboissons_index`.

Final weekly routing plan:

Vehicle	Route (stop order)	Load	Distance	Cost
VW Caddy	Crissier → Docks → Crissier	550 kg	12.79 km	7.67 CHF
Ford Transit	Crissier → Pointu → Great Escape → Doki Doki → Simplon → Loxton → La Latina → Crissier	832 kg	18.98 km	15.18 CHF
Truck	Crissier → Alloboissons → The Lacustre → Migros Ouchy → Jetée → La Rincette → Crissier	6,054 kg	164.25 km	246.37 CHF
TOTAL		7,436 kg	196.02 km	269.20 CHF

Key insight: the cost delta between the pure mathematical optimum (Scenario 3: 266.39 CHF) and the fully reality-constrained final model (Scenario 5: 269.20 CHF) is just 2.81 CHF/week, a negligible premium for a legally compliant, sustainable, and operationally sound routing plan.

The following illustrations show the roads followed by the truck (red), the Ford Traffic (yellow) and the VW Caddy (green):



4.6 Scenario Comparison Summary

Scenario	Total Cost	Total km	Truck in city?	Single model?	Feasible?
1 Unrestricted	~241 CHF*	~160 km*	Yes	Yes	X No (impractical)
2 Full exclusion	N/A	N/A	No	Yes	X No (infeasible)
3 Flexible alloc.	266.39 CHF	196.43 km	No†	Yes	✓ Yes
4 Forced lakeside	266.64 CHF	196.43 km	No†	Yes	✓ Yes
5 Final model ★	269.20 CHF	196.02 km	No†	Yes	✓ Yes RECOMMENDED

5. Sensitivity Analysis

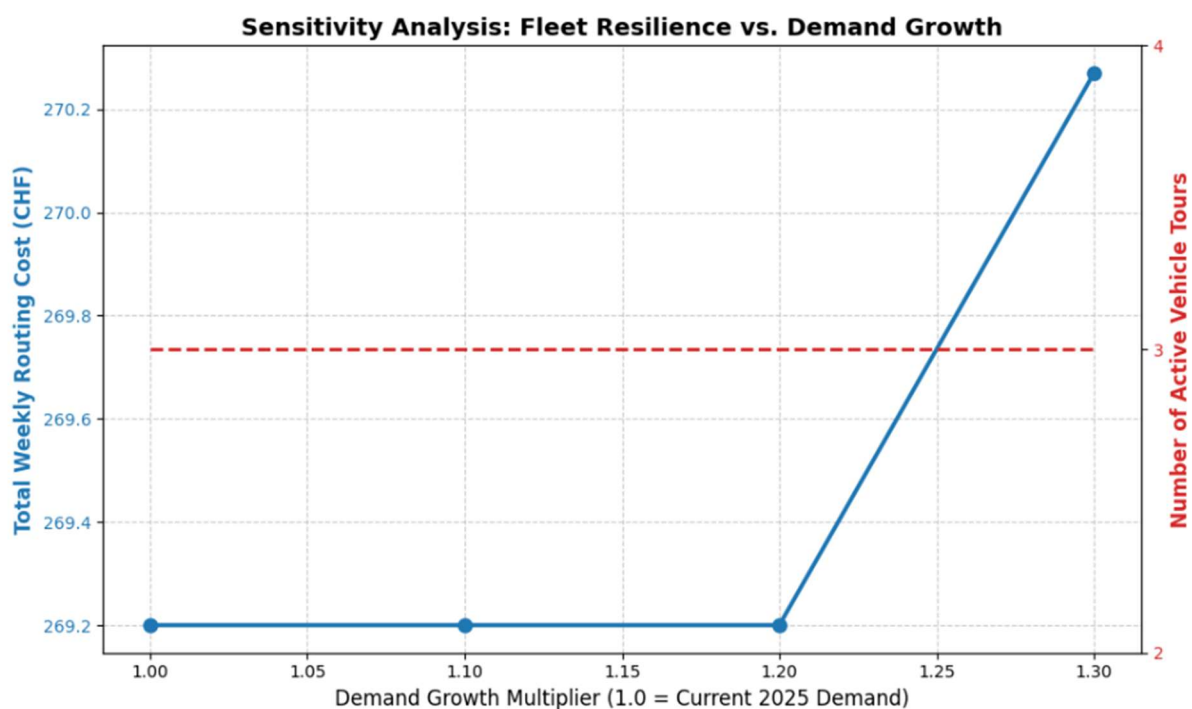
To assess the robustness and future scalability of the Scenario 5 routing plan, three sensitivity analyses were conducted: demand growth, vehicle operating cost, and CO₂ environmental impact.

5.1 Demand Growth & Fleet Resilience

In Scenario 5, the Caddy operates at 78% and the Transit at 69% capacity, leaving meaningful room for growth. The model is re-run at demand multipliers from +0% to +60% to find the tipping point at which van capacity is exhausted.

Demand Growth	Total Weekly Cost	Active Tours	Status
+0% (baseline)	269.20 CHF	3 tours	✓ Single wave
+10% to +30%	Marginal increase	3 tours	✓ Single wave
+40% to +50%	Step jump ↑	4–5 tours	△ Tour 2 activated
+60%+	High / Infeasible	-	✗ Capacity limit

Step-Cost Function: Below the tipping point (~+40–50% HORECA demand), routing costs increase only marginally. At the threshold, combined HORECA weight exceeds the 1,900 kg van capacity ceiling and a second afternoon tour activates, triggering a sudden cost jump. La Nébuleuse can absorb approximately 40% HORECA sales growth before any fleet investment is required.



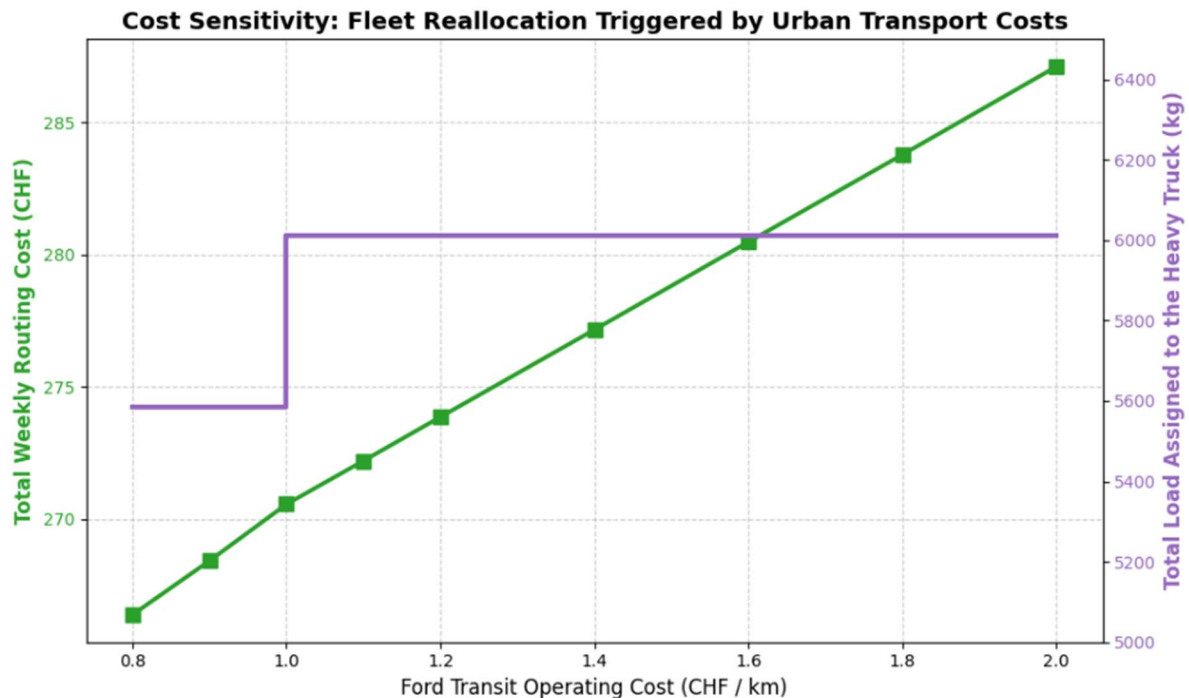
Key takeaway: La Nébuleuse has several years of operational headroom at current growth rates. The fleet should be monitored once urban weekly demand consistently approaches 1,520 kg (~80% of combined van capacity).

5.2 Vehicle Operating Cost Sensitivity

The Flexible Allocation model (Scenario 3) is re-run while increasing the Ford Transit's cost from 0.80 to 2.00 CHF/km. The solver is free to reassign the lakeside nodes to the truck or keep them with the van.

Ford Transit Cost	Total Weekly Cost	Lakeside on Truck?
0.80 CHF/km (baseline)	266.39 CHF	No (van handles lakeside)
~0.90–1.00 CHF/km	Tipping point	⚠ Solver begins to reallocate
≥1.10 CHF/km	Higher	✓ Yes (truck takes all lakeside)

Algorithmic Reallocation: At the tipping point, the solver natively offloads Jetée and La Rincette onto the truck, jumping its load from ~5,586 kg to ~6,054 kg, without any human intervention. This proves that the Scenario 5 forced assignment is mathematically justified and robust even under rising van costs.



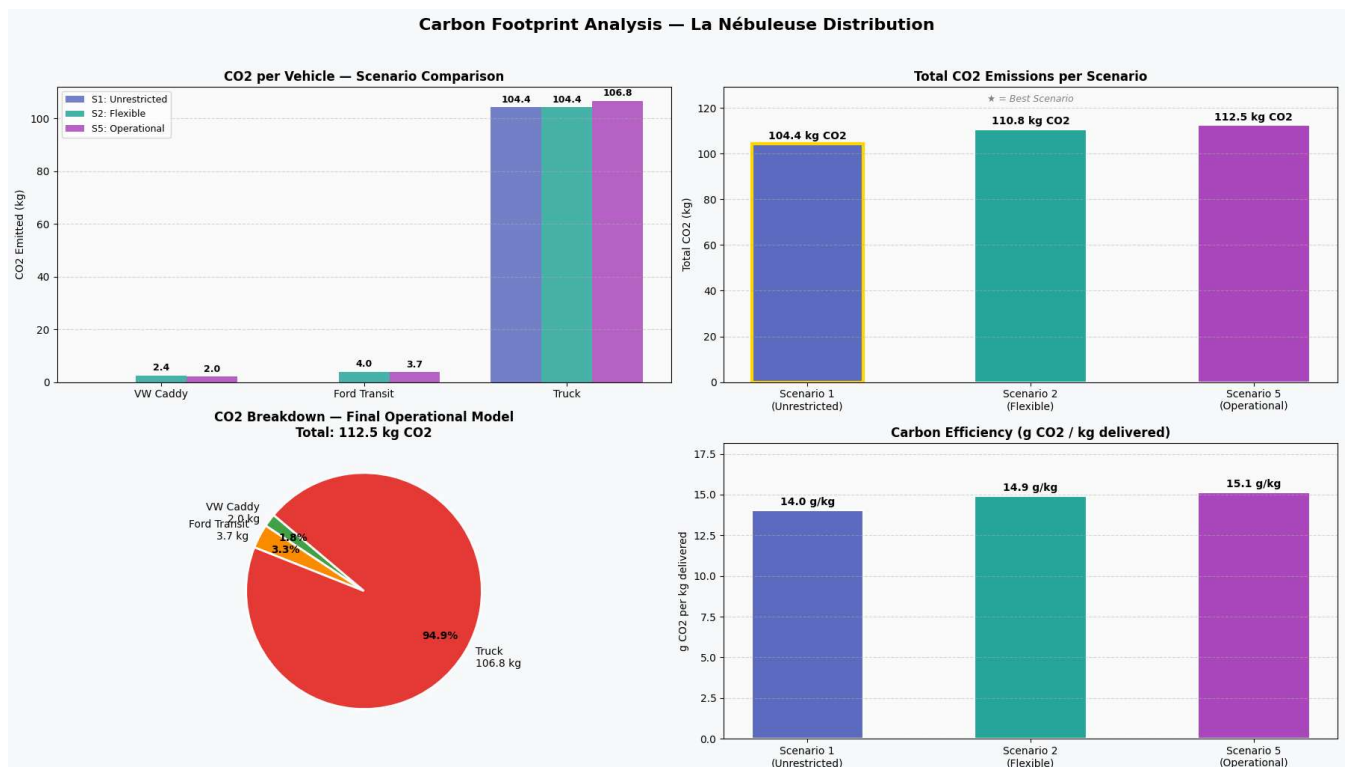
Key takeaway: The Scenario 5 plan is stable against a Ford Transit cost increase of up to ~25%. Beyond that, the model self-corrects to the same truck-lakeside assignment, the plan remains optimal regardless.

5.3 Environmental Impact & CO₂ Emissions Analysis

CO₂ emissions are calculated for three scenarios using industry-standard emission factors: VW Caddy 158 g/km, Ford Transit 195 g/km, Heavy Truck 650 g/km.

Scenario	Caddy (kg CO ₂)	Transit (kg CO ₂)	Truck (kg CO ₂)	Total CO ₂	g CO ₂ /kg delivered
S1 Unrestricted (truck only)	0.00	0.00	104.36	104.36 kg	14.0 g/kg
S2 Flexible Allocation	2.45	3.98	104.36	110.78 kg	14.9 g/kg
S5 Final Operational	2.02	3.70	106.76	112.48 kg	15.1 g/kg

- **The Heavy Truck Paradox:** Scenario 1 produces the lowest total CO₂ (104.36 kg) but is operationally impossible. At 15.1 g CO₂/kg delivered, the final model is highly efficient despite using three vehicles.
- **Cost of urban compliance:** Moving to S5 from S1 adds only 8.12 kg CO₂/week (~422 kg/year), the ecological cost of deploying appropriately-sized urban vehicles.
- **Urban fleet impact:** The Caddy and Transit together generate only 5.72 kg CO₂/week (5.1% of total). Electrifying these two vehicles would eliminate virtually all urban emissions: a strong case for EV adoption explored in Section 7.
- **Dominant source:** The truck accounts for 94.9% of total weekly emissions, almost entirely from the long-haul Givisiez run: not city driving. Urban routing choices are not the primary carbon challenge.



6. Facility Location Problem (FLP)

The VRP analysis optimised deliveries from the current two-site infrastructure. This section asks a more fundamental question: is the infrastructure itself optimal? Using a Capacitated Facility Location Problem (CFLP) formulated as a Mixed Integer Programme (MIP), we evaluate three network configurations.

6.1 Problem Definition & Infrastructure Bottlenecks

La Nébuleuse currently operates two logistical nodes:

- **Brasserie Renens (Production):** 1,200 m² brewing and packaging, ~150 pallet spaces.
- **Entrepôt Crissier (Depot):** 1,000 m² storage, ~500 pallet spaces, ~3.5 km from production.

The "First-Mile" Inefficiency: Finished goods must be constantly shuttled from the Renens brewery to the Crissier depot before dispatch. This structural separation introduces double handling, increases internal transport costs, and generates unnecessary GHG emissions before the final delivery routing even begins.

6.2 Three Strategic Scenarios

- **Scenario 1 (Baseline):** Keep current split infrastructure: production in Renens, storage/dispatch from Crissier.
- **Scenario 2 (Consolidation):** Lease adjacent space at Renens, merge operations into a single Brew-and-Dispatch facility, permanently close Crissier.
- **Scenario 3 (New Hub):** Keep Renens + Crissier, open a secondary hub in Givisiez near Alloboissons.

Key assumption: the Brasserie must remain in Renens. The FLP evaluates only the storage/dispatch decision.

6.3 Hub Candidate Data

Hub Candidate	Fixed Cost (CHF/week)	Model Capacity (pallets)	Notes
Entrepôt Crissier	1,200 CHF/week	50 pallets	Current depot (status quo)
Renens Expansion	1,000 CHF/week	50 pallets	Adjacent lease at brewery. TC1 = 0
Givisiez (new hub)	1,200 CHF/week	40 pallets	Near Alloboissons (long shuttle)

Costs estimated from Swiss commercial real estate listings using 2 m²/pallet space (benchmarked against real Crissier: 1,000 m² / 500 pallets). Demand aggregated to two clusters: Urban HORECA (4.5 pallets/week) and Alloboissons (11.0 pallets/week), 600 kg gross weight per pallet.

6.4 Mathematical Formulation (Multi-Echelon MILP)

Two-echelon structure: Tier 1 = shuttle from Renens plant to hub; Tier 2 = hub-to-customer delivery. Objective minimizes total weekly supply chain cost:

$\text{Min } Z = \sum_{(j \in J)} F_j \cdot Y_j + \sum_{(p \in P)} \sum_{(j \in J)} TC1_{pj} \cdot X1_{pj} + \sum_{(j \in J)} \sum_{(c \in C)} TC2_{jc} \cdot X2_{jc}$
Fixed facility costs + Tier 1 (plant→hub shuttle) + Tier 2 (hub→customer delivery)

Key variables:

- $Y_j \in \{0,1\}$ (hub open/closed)
- $X1_{pj} \geq 0$ (pallets shuttled plant→hub)
- $X2_{jc} \geq 0$ (pallets delivered hub→customer)

Key constraints:

- demand satisfaction
- flow conservation (hub in = hub out)
- hub capacity Big-M

6.5 Results

FLP Scenario 1: Baseline: Crissier Depot	1,332.91 CHF/week
---	------------------------------

Operating a separate Crissier facility incurs 1,200 CHF/week fixed overhead, and 100% of brewed volume must be shuttled 3.5 km simply to be staged for distribution. Total: 1,332.91 CHF/week.

FLP Scenario 2: Consolidation: Renens Expansion RECOMMENDED	1,130.65 CHF/week
--	------------------------------

Eliminating the Tier 1 shuttle ($TC1 = 0$, same coordinates) and reducing fixed cost from 1,200 to 1,000 CHF/week yields the mathematical minimum: 1,130.65 CHF/week. Savings: 202 CHF/week → over 10,500 CHF/year.

FLP Scenario 3 Dual Hub Crissier + Givisiez	2,476.32 CHF/week
--	------------------------------

Financially disastrous: 2,476.32 CHF/week, nearly double the baseline. The 1,200 CHF/week additional fixed cost of a Givisiez warehouse far exceeds the marginal savings on the Alloboissons delivery distance. Supply chains are dictated by fixed cost absorption, not variable transport savings.

Scenario	Fixed Cost (CHF/week)	Variable Cost (CHF/week)	Total Cost (CHF/week)
S1 Baseline (Crissier)	1,200	132.91	1,332.91
S2 Consolidation (Renens)	1,000	130.65	1,130.65
S3 Dual Hub	2,400	76.32	2,476.32

6.6 Managerial Interpretation & Final Recommendation

Moving tare weight (empty kegs, glass bottles, wooden pallets) between two facilities in the same metropolitan area is structural waste. Every pallet that travels from Renens to Crissier generates a cost and a carbon footprint before a single delivery has been made.

Final Strategic Recommendation: If available, La Nébuleuse should lease adjacent space at Renens and permanently close the Crissier depot, creating a single Brew-and-Dispatch facility: Zero Tier 1 waste, reduced fixed costs, and a natural EV charging hub.

7. Electric Vehicle Fleet Analysis

This section evaluates whether replacing the two diesel vans with battery-electric equivalents is financially defensible over a 10-year horizon, while quantifying lifecycle CO₂ savings under the Swiss grid (~30 g CO₂/kWh one of the cleanest in Europe). A side analysis addresses the truck, where the eActros 300 CAPEX (~350,000 CHF) makes electrification unviable today.

7.1 Parameters & Assumptions

Annual mileage from Scenario 5 VRP outputs (×52 weeks): Caddy 665 km/year | Transit 987 km/year | Truck 8,541 km/year.

Parameter	Value	Source
Diesel price	1.85 CHF/L	Avg. retail CH 2025
Electricity (commercial)	0.25 CHF/kWh	ElCom commercial tariffs
Diesel CO ₂ (well-to-wheel)	2.68 kg CO ₂ /L	EU EcoTransIT / mobitool
Swiss grid CO ₂ intensity	30 g CO ₂ /kWh	Swissgrid 2024 mix
Analysis horizon	10 years	Standard commercial vehicle ownership
Residual value at year 10	15% of CAPEX	Conservative for ICE and EV
NPV discount rate	5%	Representative Swiss SME WACC
Battery LCA CO ₂	70 kg CO ₂ /kWh	Ricardo / IVL 2023 mid-range

Vehicle	CAPEX (CHF)	Consumption	Maintenance (CHF/km)
VW Caddy (diesel)	32,000	6.5 L/100km	0.08
VW e-Caddy (EV)	45,000	19 kWh/100km	0.05
Ford Transit (diesel)	42,000	8.5 L/100km	0.10
Ford E-Transit (EV)	65,000	25 kWh/100km	0.06
Truck (diesel Actros)	85,000	28 L/100km	0.18
eActros 300 (EV)	350,000	105 kWh/100km	0.12

7.2 10-Year TCO : Urban Vans

Vehicle	Diesel 10y TCO	EV 10y TCO	Delta (Diesel-EV)	Break-even
VW Caddy	33,332 CHF	45,648 CHF	-12,317 CHF	None in 10y
Ford Transit	44,539 CHF	66,209 CHF	-21,670 CHF	None in 10y

Financial Reality: At VRP-only mileage (665 and 987 km/year), neither van reaches break-even within 10 years. The energy cost advantage (~0.05 CHF/km) cannot amortize the 13,000–23,000 CHF CAPEX premium at such low utilization. EV TCO viability is fundamentally a utilization question.

7.3 Lifecycle CO₂ (Battery-Inclusive)

Vehicle	Battery CO ₂ Debt	Diesel 10y CO ₂	EV 10y CO ₂	Net CO ₂ Saved	Carbon Payback
VW e-Caddy	3,850 kg CO ₂	1,159 kg	3,888 kg	-2,729 kg Δ	None in 10y
E-Transit	5,250 kg CO ₂	2,248 kg	5,324 kg	-3,076 kg Δ	None in 10y

Important: At VRP-only mileage, the battery manufacturing debt is so large relative to annual operational emissions (~116 and ~225 kg CO₂/year for diesel) that the EV never achieves CO₂ parity within 10 years. This is a consequence of under-utilization, not a failure of EVs.

7.4 Mileage Sensitivity — The Key Lever

Annual km	VW Caddy Diesel 10y	VW Caddy EV 10y	Caddy Delta	Caddy BE	Transit Diesel 10y	Transit EV 10y	Transit Delta	Transit BE
1,000	34,002	45,975	-11,972	-	44,572	66,225	-21,652	-
5,000	42,012	49,875	-7,862	-	54,862	71,125	-6,262	-
10,000	52,025	54,750	-2,725	-	67,725	77,250	-9,525	-
15,000	62,038	59,625	+2,412	Yr 9	80,588	83,375	-2,788	-
20,000	72,050	64,500	+7,550	Yr 7	93,450	89,500	+3,950	Yr 9

Key finding: Financial break-even is reached at ~15,000 km/year for the Caddy (year 9) and ~20,000 km/year for the Transit. These thresholds are realistic once all van duties (supplier runs, festivals, ad-hoc) are counted.

7.5 Side Analysis : Heavy Truck

Metric	Diesel Truck	eActros 300
CAPEX	85,000 CHF	350,000 CHF
CAPEX premium	-	+265,000 CHF
Annual opex	~5,962 CHF/yr	~3,267 CHF/yr
10y TCO	144,616 CHF	382,669 CHF
10y TCO delta	-	-238,053 CHF (EV far more expensive)
TCO break-even	-	None within 10 years
Battery CO ₂ debt	-	23,520 kg CO ₂
10y CO ₂ saved	-	+37,881 kg
Carbon payback	-	Year 4

Truck conclusion: Defer eTruck. The 265,000 CHF CAPEX premium makes the truck EV 238,053 CHF more expensive over 10 years: no financial break-even within the ownership horizon. The carbon case is strong (payback year 4, 37,881 kg CO₂ saved over 10y), but cannot override the financial deficit. Revisit in 2028-2030. Evaluate HVO biodiesel as a transitional decarbonisation lever (~85% lifecycle CO₂ reduction, drop-in compatible).

7.6 Final EV Recommendation

Urban vans: Conditional electrification: Audit actual van mileage via telematics or fuel-card data before committing. If real-world km exceeds 8,000-10,000/year (likely once all duties are counted), proceed with electrification at the next replacement cycle. The carbon case stands independently even below financial break-even.

Bottom line: Electrify the urban fleet if utilization data confirms more than VRP-only duties. Defer the truck pending market evolution. Treat the carbon case as the primary justification regardless of TCO outcome. Consolidating facilities at Renens (Section 6) creates a natural single charging hub, making the EV and FLP recommendations mutually reinforcing.

8. Recommendations

This section consolidates all managerial recommendations from the VRP (Sections 3-5), FLP (Section 6), and EV analysis (Section 7), ordered by implementation priority.

#	Recommendation	Key Metric	Priority
R1	Deploy Scenario 5 as the standard weekly routing plan	269.20 CHF/week 196.02 km	Immediate
R2	Consolidate facilities: close Crissier, expand Renens	Save ~202 CHF/week (~10,500 CHF/year)	Short-term
R3	Audit actual van mileage before EV investment	Break-even at ~15,000 km/year (Caddy)	Short-term
R4	Monitor HORECA demand against the +40% threshold	Fleet viable up to +40% growth	Ongoing
R5	Consider EV vans at next replacement if km > 8,000/yr	Carbon payback under realistic utilization	Medium-term
R6	Defer eTruck — revisit 2028–2030	238,053 CHF more expensive over 10y	Deferred
R7	Explore HVO biodiesel for truck decarbonisation	~85% lifecycle CO ₂ reduction vs diesel	Medium-term

R1 Deploy Scenario 5 as the Standard Weekly Routing Plan

The Scenario 5 model should become the operational standard. Total: 269.20 CHF/week | 196.02 km. Routes:

- **VW Caddy:** Crissier → Docks → Crissier (550 kg / 700 kg = 79%)
- **Ford Transit:** Crissier → Pointu → Great Escape → Doki Doki → Simplon → Loxton → La Latina → Crissier (832 kg / 1,200 kg = 69%)
- **Truck:** Crissier → Alloboissons → The Lacustre → Migros Ouchy → Jetée → La Rincette → Crissier (6,054 kg / 8,300 kg = 73%)

R2 Consolidate Facilities: Close Crissier, Expand Renens

The FLP analysis identifies 202 CHF/week of structural waste from inter-facility shuttling. FLP Scenario 2 (1,130.65 CHF/week) eliminates this entirely: zero Tier 1 transport, reduced fixed overhead, shared staff and equipment, and a natural EV charging hub. However, it is not always feasible to find suitable rental spaces. If such an opportunity arises, La Nébuleuse should consider it. It should also be noted that the results presented do not take into account potential differences in rental costs between locations.

R3/R4/R5/R6/R7 Fleet & Growth Management

R3 Mileage audit (short-term): Install telematics or review fuel-card data. If real-world km exceeds 8,000/year, the financial case for EV becomes viable within the ownership horizon.

R4 Demand monitoring (ongoing): Track weekly urban delivery weight. When it consistently approaches 1,520 kg (~80% of combined van capacity), proactively plan for a second tour or additional vehicle.

R5 EV vans (medium-term): Consider at next replacement cycle if utilization confirms >8,000 km/year. The Swiss grid makes operational EV emissions near-zero regardless of financial outcome.

R6 Truck EV (deferred): Do not electrify the truck under current market conditions. The 265,000 CHF CAPEX premium cannot be recovered within 10 years at 8,541 km/year. Revisit in 2028–2030.

R7 HVO Biodiesel (medium-term): As a transitional measure, Hydrotreated Vegetable Oil offers ~85% lifecycle CO₂ reduction vs diesel, is drop-in compatible, and is increasingly available at Swiss commercial stations.

9. Conclusion & Future Extensions

9.1 Summary

This project applied three complementary operations research methods to La Nébuleuse's logistics network, yielding actionable, data-driven recommendations across routing, facility strategy, and fleet electrification.

The HFVRP demonstrated that pure mathematical optimization produces an unworkable plan. By progressively introducing operational constraints across five scenarios, the final model achieves a routing plan that is legally compliant, environmentally responsible, and financially near-optimal, at a cost premium of less than 3 CHF/week over the unconstrained baseline.

The FLP revealed a structural first-mile inefficiency: 202 CHF/week is spent shuttling finished goods between Renens and Crissier. Consolidating into a unified Brew-and-Dispatch facility at Renens is the financially dominant strategy, saving over 10,500 CHF/year.

The EV analysis provides an honest verdict: at VRP-only mileage, the financial case does not close within 10 years. However, the carbon case is robust under realistic utilization, and financial break-even is achievable once all van duties are counted. The truck should not be electrified under current market conditions.

Bottom line: Electrify the urban fleet if utilization data confirms more than VRP-only duties. Consolidate facilities at Renens. Deploy Scenario 5 routing immediately. Defer the eTruck pending market evolution. Treat the carbon case as the primary justification regardless of TCO outcome.

9.2 Future Extensions

Several natural extensions would further enhance the model's real-world value:

- **Time Windows:** Incorporate delivery time windows (bars 07:00-11:00, restaurants after 14:00) using OR-Tools' time dimension module. This would also handle the 5% of urgent next-day deliveries.
 - **Stochastic Demand Modelling:** Replace deterministic weekly averages with stochastic VRP techniques, sampling demand from historical distributions to generate robust routes across a range of demand realizations.
 - **Multi-Period FLP:** Model the transition cost of closing Crissier and expanding Renens over time (lease break penalties, moving costs, construction timelines) to identify the optimal transition moment.
 - **CO₂-Weighted Objective Function:** Replace or complement cost minimization with a weighted emission objective, making the sustainability dimension explicit and quantifiable within the optimization itself.
 - **Real-Time Re-Optimisation:** Integrate the OR-Tools model with La Nébuleuse's ERP to automatically re-optimize routes each Monday morning based on confirmed weekly orders.
 - **Fleet Sizing Optimisation:** Co-optimize the number and type of vehicles alongside routing decisions. Answering: given projected demand growth, what is the optimal fleet composition for the next 5 years?
 - **Multi-Depot Extension:** If the FLP Scenario 2 consolidation is adopted, update the routing model to a single-depot configuration. If any secondary hub is ever opened, extend to a full multi-depot VRP.
-